

Printable solar materials are one of the examples that could turn parts of a house into solar panels. Quantum dots are set to revolutionise the field of displays, improving resolution and colour contrast. Nanosensors for agricultural pest management are already on their way to the market.

Virtual assistants

Personal digital assistants like Amazon Alexa are already present in the market, and many more will be released this year.

An automated grocery store is a good example of virtual assistants being deployed on a large scale. This is eliminating the need for cashiers. Watasale, India's first cashier-less store in Kochi, Kerala, is a major tech triumph of 2018.

Virtual assistants allow users to control anything through voice commands, and perform services or tasks on their behalf. These exploit AI, NLP, ML, voice processing, and reasoning and knowledge representation to make human-machine interactions simpler, more natural and appealing.

Chatbots are shaping the business

growth story. By 2020, the technology will reach its full market potential, with majority of businesses employing chatbots to redefine their customer engagement policies with the integration of NLP.

AR and VR

AR and VR will revolutionise the construction industry and change the way buildings are made. This will help with planning and plotting sites before even breaking ground.

AR tools help healthcare workers save lives. AR and VR will be used extensively in training and guiding doctors and the military.

VR has revolutionised the gaming industry, too.

AR is playing a vital role in Industry 4.0. It helps select manufacturing parts and send industrial fault status information to applications installed on users' phones and assemblies. Many companies are developing applications to provide real-time information about product processing to workers. They are also making smart applications that

help with decision-making and improved work procedures.

To sum up

The technology today has world-changing potential in the near future. By 2020, the human face will become the ideal technology partner, from unlocking a phone, house or car, to withdrawing money from a bank ATM.

Apart from giving data related to how users will react, technology will also guide businesses on how they should act when users behave in a certain way. Hence, companies will be investing more in educating and training workforces for the coming digital era.

Several technological revolutions will come and go. What is important is to stay updated with all the latest developments and to understand how these will affect the world. **END** 

The article was originally published in the January 2019 issue of *Electronics For You*.

 By: Deepshikha Shukla

The author is a technology journalist at EFY.

Continued from Page...31

It also paves the way for new digital services to be rapidly built, changed, enhanced and deployed, so that new ideas for greater transparency, effectiveness and efficiency can be tried and refitted, if needed. Open source helps the enterprise application architecture to be technology independent and provides openness and transparency. It makes enterprise data available openly, so that external parties can build services, with adequate visibility of how the enterprise functions.

From the statistics, we can clearly make out that PaaS adoption is picking up very rapidly. Enterprises can now see the benefits PaaS provides by lowering capex and opex. PaaS enables the enterprise application to be run on any cloud platform, whether public or private. **END** 

References

- [1] <https://www.idc.com/getdoc.jsp?containerId=US43171317>
- [2] <https://www.forbes.com/sites/forbestechcouncil/2018/07/05/four-trends-in-cloud-computing-cios-should-prepare-for-in-2019/#69125c2c4dc2>
- [3] https://www.pointivity.com/latest_technology_trends
- [4] <https://www.gartner.com/newsroom/id/3871416>
- [5] <https://erpinnews.com/enterprise-workloads-will-be-in-the-cloud>

Acknowledgements

The author would like to thank Hari Kishan Burle and Raju Alluri of the Global Enterprise Architecture Group of Wipro Technologies for giving their time and support to bring out this article as part of the Architecture Practices effort.

Disclaimer

The views expressed in this article are that of author's and Wipro does not subscribe to the substance, veracity or truthfulness of the said opinions.

 By: Dr Gopala Krishna Behara

The author is a lead enterprise architect in the Global Enterprise Architecture division of Wipro. He has 21 years of IT experience. He can be reached at gopalkrishna.behra@wipro.com.